Response to Reviewers

Manuscript: "Empirical Assessment of Storm-time Thermospheric Density Inversion Methods from LEO POD Data"

Date: May 2026

We thank the Editor and both Reviewers for the continued careful re-review of the revised manuscript. Below we address each comment in turn, quoting the original reviewer text and describing the change made. Where applicable, we reproduce the relevant old and new manuscript text to demonstrate the specific revisions.

Throughout this response:

- "OLD" refers to the previously submitted (Round 1 revised) manuscript.
- "NEW" refers to the manuscript as resubmitted in this revision.
- Section, equation, table, and figure numbers refer to the revised manuscript.

Reviewer 2 Comments

Comment R2-47

"Is there a reference to back this statement? Can you show the PDF of the Thermospheric densities *to justify this statement?" (Section 2.4: "Thermospheric densities are approximately log-normally distributed (Emmert, 2015; Picone et al., 2005)")*

Response: The decision to adopt log-normal evaluation metrics was made in Round 1 at the explicit recommendation of Reviewer 1, who wrote:

"the use of log-normal metrics is highly recommended and consistent with recent literature on thermospheric data model comparisons."

To strengthen the supporting reference at the cited sentence we add Fitzpatrick et al. (2025), which states directly:

"The use of logarithms also ties into the nature of TMD, which displays an approximately log-normal distribution at a fixed altitude. Applying the logarithm to the ratio of two log-normally distributed positive-definite variables transforms the result into a normal distribution, streamlining both statistical analysis and interpretation."

The Emmert (2015) reference is retained for the broader physical motivation (log-density variance is more uniform than density variance; Emmert and Picone, 2010, §2.2). We have removed the Picone (2005) citation at this specific sentence, since that paper does not contain an explicit log-normality statement.

Manuscript change in §2.4:

OLD:

"Thermospheric densities are approximately log-normally distributed (Emmert, 2015; Picone et al., 2005), so the natural variable for assessing retrieval accuracy is $\ln \rho$ rather than ρ."

NEW:

"Thermospheric densities are approximately log-normally distributed at a given altitude (Emmert, 2015; Fitzpatrick et al., 2025); the variance of $\ln \rho$ is also far more uniform than that of ρ across solar cycle and storm states (Emmert and Picone, 2010, §2.2). The natural variable for assessing retrieval accuracy is therefore $\ln \rho$ rather than ρ."

Comment R2-64 / R2-82: Significance of the EDR advantage

"Yes, but the effect seems to be small. Is this significant?" (Section 3.2: "EDR outperforms POD-A on every metric")

Response: We agree that, with the EDR vs POD-A differences being modest, the reader is right to ask whether they are statistically meaningful. We have added 95 % confidence intervals on the matched-reference metrics in Table 3 directly into the manuscript text in §3.2. The interval methods are chosen to be exact under the log-normality of the residuals already established in §2.4. Specifically: a chi-squared interval on σ, because the sample variance of normally distributed log-residuals follows the chi-squared distribution with $n-1$ degrees of freedom; a chi-squared interval on $\tilde{\rho}^2_{\mathrm{RMS}}$, by the same argument applied to the squared %-error sum; and a Fisher z-transform interval on r^2, the standard inferential interval for a Pearson correlation under bivariate normality of the (log-)data.

The intervals reveal that the answer depends on the metric. At 1-orbit cadence, $\sigma_{\mathrm{POD-A}} = 0.115$ [0.112, 0.118] and $\sigma_{\mathrm{EDR}} = 0.116$ [0.113, 0.119] (the intervals overlap, so the methods are statistically indistinguishable in typical-orbit scatter), whereas r^2 (0.985 [0.984, 0.986] vs 0.992 [0.991, 0.992]) and $\tilde{\rho}_{\mathrm{RMS}}$ (13.5 % [13.2, 13.8] vs 12.0 % [11.8, 12.3]) differ significantly (the intervals do not overlap).

The qualitative interpretation is that EDR and POD-A produce similarly-scattered errors on a typical orbit (so σ does not distinguish them), but EDR tracks the time-variation of the accelerometer truth more faithfully (so r^2 is higher) and produces fewer catastrophic-failure orbits ($\tilde{\rho}_{\mathrm{RMS}}$ squares the per-orbit %-error, so worst-case orbits dominate the pooled statistic). This is consistent with the robustness advantage discussed elsewhere in §3.3.

Manuscript change in §3.2 (inserted between the existing "EDR outperforms POD-A…" sentence and the "matched-to-native jump…" sentence):

NEW:

"95 % confidence intervals on the matched-reference metrics are exact under the log-normality of the residuals (§2.4): the sample variance of normally distributed log-residuals follows a chi-squared distribution with $n-1$ degrees of freedom (used here to interval-bound σ and $\tilde{\rho}^2_{\mathrm{RMS}}$), and the Fisher z-transform provides the corresponding inferential interval for the Pearson correlation underlying r^2. Applying these to the matched 1-orbit row, σ is indistinguishable between methods ($\sigma_{\mathrm{POD-A}} = 0.115$ [0.112, 0.118] vs $\sigma_{\mathrm{EDR}} = 0.116$ [0.113, 0.119]), whereas r^2 and $\tilde{\rho}_{\mathrm{RMS}}$ differ significantly ($r^2 = 0.985$ [0.984, 0.986] vs 0.992 [0.991, 0.992]; $\tilde{\rho}_{\mathrm{RMS}} = 13.5\,\%$ [13.2, 13.8] vs 12.0 % [11.8, 12.3]). At single-orbit cadence the two methods produce similarly-scattered errors on a typical orbit (equivalent σ), but EDR follows the time-variation of the accelerometer truth more faithfully (higher r^2) and produces fewer catastrophic-error orbits (lower $\tilde{\rho}_{\mathrm{RMS}}$, since $\tilde{\rho}_{\mathrm{RMS}}$ squares the %-error and is dominated by the worst orbits)."

Comment R2-82 is the conclusions-section companion to R2-64. We have softened the wording in the Q3 paragraph of the Conclusions correspondingly, replacing "outperforms POD-A on every metric" with language that distinguishes between metrics where the EDR advantage is significant (r^2, $\tilde{\rho}_{\mathrm{RMS}}$) and those where the methods are statistically indistinguishable (σ at 1- and 2-orbit cadence).